

Chapter 12: Nonequilibrium (Quantum):

Exercise #1 (pg 699) "Projector"

(pg 701)

$$P \equiv \hat{\rho}_b^{eq} + \text{tr}_b(\cdot)$$

(pg 700) "Short hand notation"

$$\hat{\rho}_P = P \hat{\rho} \quad \hat{\rho}_Q = Q \hat{\rho}$$

(pg 699) "Complementary Projection"

$$Q \equiv id - P$$

(12.1) "Projector Equations"

$$P \hat{L}_b = L_b P = 0$$

$$[\hat{L}_s, P] = 0$$

$$P \hat{L}_i P = 0$$

where L_b = "Liouville operator"

L_s = "Liouville operator For system"

L_i = "Liouville coupling"

(Pg 701) "Relationship about Integral Kernel"

$$P \hat{L}_i e^{t' Q L_0 Q} L_i \hat{\rho}_P(t) = P \hat{L}_i e^{t' L_0} L_i \hat{\rho}_P(t)$$

because $Q=1$ is the root to projector equation, $P \hat{L}_i P = (1-Q) \hat{L}_i (1-Q) = 0$

Exercise #2: IF $t \ll 1$, then $e^{t L_0} \hat{O} = \hat{O}(t) e^{t L_0}$

(Pg 701)

Exercise #3: (12.5) "Markovian Quantum Master Equation"

(Pg 703)

$$\begin{aligned} \partial_t \hat{\rho}_s &= (\hat{L}_s + \alpha^2 \int_0^t dt' \langle L_0(0) L_i(-t') \rangle_b) \hat{\rho}_s \\ &= \hat{L}_s \hat{\rho}_s + \alpha^2 \int_0^t dt \langle (-i [\hat{H}_{s,b,i}, \cdot]) \rangle_b \hat{\rho}_s \end{aligned}$$

(Pg 699)

"Quantum Liouville Operator"

$$\hat{L}_{s,b,i} \equiv -i [\hat{H}_{s,b,i}, \cdot]$$

With pg 701, "System-bath coupling",

$$\hat{H}_i = \sum_n (c_n \hat{T}_n \otimes \hat{X}_n + \bar{c}_n \hat{T}_n^\dagger \otimes \hat{X}_n^\dagger)$$

$$\partial_t \rho_s = \hat{L}_s \hat{\rho}_s + \alpha^2 \int_0^t dt' \left(\sum_n c_n \hat{T}_n \otimes \hat{X}_n + \bar{c}_n \hat{T}_n \otimes \hat{X}_n^\dagger \right)^2 \hat{\rho}_s$$

$$= \hat{L}_s \hat{\rho}_s + \left(\sum_n c_n \hat{T}_n \otimes \hat{X}_n + \bar{c}_n \hat{T}_n \otimes \hat{X}_n^\dagger \right)^2 \hat{\rho}_s$$

↑
Pg 701, "c_n are coupling

constants defined to
absorb the global
constant α above"

$$= \hat{L}_s \hat{\rho}_s + \sum_n \left(c_n^2 \hat{T}_n \otimes \hat{X}_n \cdot \hat{T}_n \otimes \hat{X}_n + c_n \bar{c}_n \hat{T}_n \otimes \hat{X}_n \cdot \hat{T}_n \otimes \hat{X}_n^\dagger + \bar{c}_n c_n \hat{T}_n \otimes \hat{X}_n^\dagger \cdot \hat{T}_n \otimes \hat{X}_n + c_n^2 \hat{T}_n^\dagger \otimes \hat{X}_n^\dagger \cdot \hat{T}_n^\dagger \otimes \hat{X}_n^\dagger \right) \hat{\rho}_s$$

Also from pg 701, "Bath Memory"

$$\langle T_n(0) T_n^\dagger(t) \rangle_b = K_n^\dagger \delta(t)$$

$$\langle T_n^\dagger(0) T_n(t) \rangle_b = K_n \delta(t)$$

$$\begin{aligned}\partial_t \rho_s &= \hat{L}_s \hat{\rho}_s + \sum_n (|c_n|^2 \hat{X}_n^\dagger + c_n^\dagger |c_n|^2 \hat{X}_n \hat{\rho}_s \hat{X}_n^\dagger) \\ &= \hat{L}_s \hat{\rho}_s + \sum_n \left[\gamma_n^\dagger (2 \hat{X}_n \hat{\rho}_s \hat{X}_n - \{ \hat{X}_n \hat{X}_n^\dagger, \rho_s \}) \right. \\ &\quad \left. + \gamma_n (2 \hat{X}_n^\dagger \rho_s \hat{X}_n - \{ \hat{X}_n^\dagger \hat{X}_n, \rho_s \}) \right]\end{aligned}$$

Exercise #4: (pg 703) "Linblad equation-modified"

(pg 703)

$$\int_0^t \langle \hat{L}_i(0) \hat{L}_i(-t') \rangle_0 \rho_s = - \sum_k |c_k|^2 \int_0^\infty dt'$$

$$\times (e^{-i(\omega_k - \epsilon)t'} (\langle \hat{n}_k \rangle (\hat{\rho}_s a a^\dagger - a^\dagger \rho_s a) + \langle \hat{n}_k + 1 \rangle (a a^\dagger \hat{\rho}_s - \hat{\rho}_s a^\dagger)))$$

$$\times e^{i(\omega_k - \epsilon)t'} (\langle \hat{n}_k \rangle (\hat{\rho}_s^\dagger a a - a^\dagger \hat{\rho}_s^\dagger a) + \langle \hat{n}_k + 1 \rangle (\hat{\rho}_s^\dagger a a - a^\dagger \rho_s^\dagger a))$$

$$\begin{aligned}&= \pi \sum_k |c_k|^2 \delta(\epsilon - \omega_k) \times (\langle n_k \rangle (2 a^\dagger \rho_s a - \hat{\rho}_s a a^\dagger - a a^\dagger \rho_s) \\ &\quad + (\langle n_k + 1 \rangle (2 a \hat{\rho}_s^\dagger - \rho_s^\dagger a^\dagger - a^\dagger a \rho_s)))\end{aligned}$$

The author (Atland) asked about $[\rho_s, a a^\dagger] = 0$, specifically the commutation: " $\rho_s a a^\dagger - a a^\dagger \rho_s$ ".

Mr. Atland put zero-value terms into the lines above.

$$= \pi \sum_k |c_k|^2 \delta(\epsilon - \omega_k) \times (\langle n_k \rangle \circ 2 a^\dagger \rho_s a + \langle n_k + 1 \rangle \circ 2 a \hat{\rho}_s^\dagger a^\dagger)$$

$$= |c_\omega|^2 \rho(\omega) \langle n_\omega \rangle \circ a^\dagger \hat{\rho}_s a + \langle n_k + 1 \rangle \circ a \hat{\rho}_s^\dagger a^\dagger$$

$$= P \int d\omega \frac{|c\omega|^2 \rho(\omega)}{\omega - E} \langle n_\omega \rangle [a^\dagger a, \rho_s]$$

$\uparrow$
P is "Principal value Integral"

from Cauchy i.e. a limit.

The final result describes difference between frequency with a set of distributions.

Exercise #5: (12.12) "Coherent states"

(pg 708)

$$\langle \psi | e^{c a^\dagger} | \psi \rangle = \int_{-\infty}^{\infty} \frac{c}{2\pi} e^{c a^\dagger} \psi d\tau$$

$$= e^{c a^\dagger} \int_{-\infty}^{\infty} \frac{c}{2\pi} \psi d\tau$$

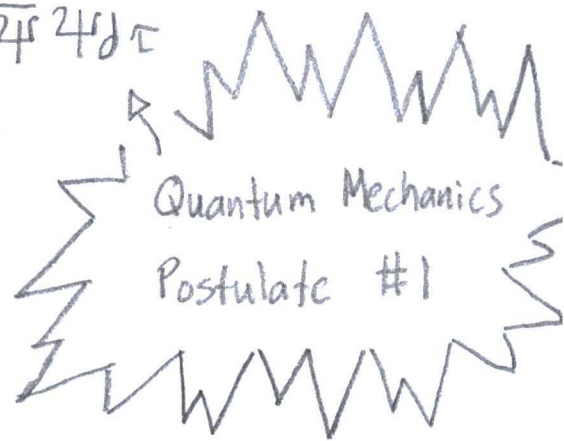
$$= e^{c a^\dagger}$$

$$= 1 @ c=0$$

$$e^{c \frac{d}{d\tau}} = e^{\frac{d}{d\tau} \int_{-\infty}^{\infty} d\tau e^{c \frac{d}{d\tau}} \psi}$$

$$= e^{\frac{d}{d\tau} \int_{-\infty}^{\infty} d\tau \psi} @ c=0$$

$$= 1$$



Exercise #6:
(pg 709)

(pg 709) "Matrix Property"

$$\det(-iG^{-1}) = \det(-i \cdot i \begin{bmatrix} -a_+ & 1 & & \\ & \ddots & \ddots & \\ & & -a_+ & 1 \\ -1 & & & -a_- & 1 & \\ & & & & \ddots & \ddots \\ & & & & & -a_- & 1 \end{bmatrix} - K)$$

First time
in all life
(Equation 12.14)

$$= \exp \left[\text{tr} \left[\ln \begin{bmatrix} -a_+ & 1 & & -K \\ & \ddots & \ddots & \\ & & -a_+ & 1 \\ -1 & & & -a_- & 1 & \\ & & & & \ddots & \ddots \\ & & & & & -a_- & 1 \end{bmatrix} \right] \right] \quad \text{where } a_{\pm} = 1 \mp i\epsilon$$

$$= \exp \left[\text{tr} \left[- \left(\begin{bmatrix} + & & \\ & + & \\ & & + \end{bmatrix} - 1 \right) - \frac{([+]-1)^2}{2} - \dots \right] \right]$$

<u>Logarithm Series</u> $\log(1-x) = -x - \frac{x^2}{2!} - \dots$
--

$$= 1 + \left[- \left(\begin{bmatrix} + & & \\ & + & \\ & & + \end{bmatrix} - 1 \right) - \dots \right]$$

$$= 1 - k(a_+ a_-)^{N-1} - 1 \cdot (a_+ a_-) + 1$$

$$= 1 - k(a_+ a_-)^{N-1}$$

Exercise #7:
(pg 709)

(From exercise) "Matrix Hamiltonian"

$$M = \mathbb{1} - H$$

$$M^{-1} = (\mathbb{1} - H)^{-1}$$

$$= 1 + H + H^2 + H^3 + \dots$$

Exercise #8:
(pg 729)

$$\hat{\Sigma}_b = \begin{pmatrix} \hat{\Sigma}_b & \hat{\Sigma}_b^K \\ 0 & \hat{\Sigma}_b \end{pmatrix}$$

where $\hat{\Sigma}_b^\pm(x, x') = -\frac{i}{2} (G^\mp(x', x) G^K(x, x') + G^\pm(x, x') G^K(x', x))$

$$\hat{\Sigma}_b^K(x, x') = -\frac{i}{2} (G^K(x, x') D^K(x, x') + (G^+ - G^-)(x, x') \cdot (D^+ - D^-)(x, x'))$$

Exercise #9: (A.57) "Spatio-temporal Wigner Transform"
(pg 731)

$$X(x_1, x_2) = \int dp e^{-i(x_1 - x_2)^T \eta p} X(x, p)$$

with $\eta = \text{"Minkowski Metric"}$

(from exercise) "Operator $\hat{Z}$ "

$$Z(x, x') = X(x, x') Y(x, x')$$

$$= \int dp e^{-i(x - x')^T \eta p} X(x, x') Y(p - p')$$

$$= \int dp' X(p') Y(p - p') \quad @ x = 0$$

$$Z(x, x') = X(x, x') Y(x', x)$$

= ...

$$= \int dp' X(p') Y(p'-p)$$

Exercise #10:
(pg 735)

(Chapter 5) "Hubbard-Stratonovich Transformation"

$$e^{-\frac{a}{2}x^2} = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2a} - ixy} dy$$

(12.83) "Hamiltonian for Quantum Dots"

$$\hat{H}_a = \sum_{ab} C_{\sigma,a}^\dagger H_{\sigma,ab} C_{\sigma,b}$$

$$\hat{H} = \sum_{\sigma, \mu} C_{\sigma,\mu}^\dagger T_{\sigma,\mu} d_\mu + c$$

$$\hat{H}_d = \sum_{\mu\nu} d_\mu^\dagger H_{d,\mu\nu} d_\nu$$

$$\hat{H}_c = \frac{E_c}{2} \left(\sum_{\mu} d_\mu^\dagger d_\mu - N_0 \right)^2$$

(B.71) "Exponent from 3.71"

$$-\frac{i}{\hbar} S[q, \dot{q}] = -\frac{i}{\hbar} \int_0^t L(q, \dot{q}) dt$$

$$= -\frac{i}{\hbar} \int_0^t \left(\frac{1}{2} m \dot{q}^2 - V(q) \right) dt$$

$$= -\frac{i}{\hbar} \int_0^t \frac{1}{2} p \cdot \dot{q} - H(x, p) dt$$

$$S[q, \dot{q}] = \int_0^t \frac{1}{2} p \cdot \dot{q} - H(x, p) dt$$

$$S[\psi, V] = \sum_{\sigma=L, R} S_{\sigma}[\psi_{\sigma}] + S_{t, \sigma}[\psi_{\sigma}^L, \psi_{\sigma}^R] + S_d[\psi_d, V] + S_c[V]$$

$$= \sum_{\sigma=L, R} \int H_{\sigma}[\psi_{\sigma}] dt + \int H_{t, \sigma}[\psi_{\sigma}^L, \psi_{\sigma}^R] dt + \int H_d[\psi_d, V] dt + \int H_c[V] dt$$

$$= \sum_{\sigma=L, R} \int \psi_{\sigma}^{\dagger} \begin{pmatrix} i\partial_t + E_F^{\dagger} - H & 2i\delta F \\ 0 & i\partial_t + E_F - H \end{pmatrix} \psi_{\sigma} dt$$

$$+ \int \left(\bar{\psi}_{\sigma} \begin{pmatrix} \hat{T}_{\sigma} & 0 \\ 0 & \hat{T}_{\sigma}^{\dagger} \end{pmatrix} \psi_d + \bar{\psi}_d \begin{pmatrix} \hat{T}_{\sigma}^{\dagger} & 0 \\ 0 & \hat{T}_{\sigma} \end{pmatrix} \psi_{\sigma} \right) dt$$

$$+ \int \bar{\psi}_d \begin{pmatrix} i\partial_t + E_F^{\dagger} - H_d - V_c & -V_q/2 + 2i\delta F_d \\ -V_q/2 & i\partial_t + E_F - H_d - V_c \end{pmatrix} \psi_d dt$$

$$+ \int \left(\frac{1}{2E_c} V_F V_q + N_0 V_q \right) dt$$

$$\text{when } V_c = \frac{1}{2}(V_+ + V_-)$$

$$V_q = V_+ - V_-$$

Exercise #11:

(pg 736)

Exercise #12:
(pg 739)

(12.95) "Ansatz for Left and Right
Quantum Dot Function"

$$F_d = \frac{g_L F_L + g_R F_R}{g_L + g_R}$$

(12.87) "Tunneling Action"

$$S_{\text{tun}}[\Phi] = -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} \left(\Lambda_{\sigma} e^{-i\hat{\Phi}} \Lambda_d e^{i\hat{\Phi}} \right)$$

$$\text{Where } g_{\sigma} = 4\pi^2 v_d v_{\sigma} |T_{\sigma}|^2$$

$$= -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} \left(\Lambda_{\sigma} \left(1 - i\Phi - \frac{\Phi^2}{2} \right) \right. \\ \left. \times \Lambda_d \left(1 + i\Phi - \frac{\Phi^2}{2} \right) \right)$$

$$\text{where } \Lambda_{\sigma,d} = 2i \begin{pmatrix} g^+ & g^k \\ 0 & g^- \end{pmatrix} \epsilon$$

$$F = -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} [\Lambda_{\sigma}, \Lambda_{\sigma}] \hat{\Phi} = 0$$

= 0, because real value
solutions.

Commutator:

(12.89) "Matrix with energy dependence"

$$\underline{\Lambda}_x \equiv 2i \begin{pmatrix} g^+ & g_x^k \\ 0 & g^- \end{pmatrix} \epsilon$$

$$g_x^k = (g^+ - g^-) F_x$$

$$x = d, \sigma$$

$$[\underline{\Lambda}_d, \underline{\Lambda}_\sigma] = \underline{\Lambda}_d \underline{\Lambda}_\sigma - \underline{\Lambda}_\sigma \underline{\Lambda}_d$$

$$= 2i \begin{pmatrix} g^+ & g_d^k \\ 0 & g^- \end{pmatrix} \begin{pmatrix} g^+ & g_\sigma^k \\ 0 & g^- \end{pmatrix} - 2i \begin{pmatrix} g^+ & g_\sigma^k \\ 0 & g^- \end{pmatrix} \begin{pmatrix} g^+ & g_d^k \\ 0 & g^- \end{pmatrix}$$

$$= -4 \begin{pmatrix} (g^+)^2 & g^+ g_\sigma^k + g_d^k g^- \\ 0 & (g^-)^2 \end{pmatrix} + 4 \begin{pmatrix} (g^+)^2 & g^+ g_d^k + g_\sigma^k g^- \\ 0 & (g^-)^2 \end{pmatrix}$$

$$= -4 \begin{pmatrix} (g^+)^2 & g^+(g^+ - g^-)F_\sigma + (g^+ - g^-)F_d g^- \\ 0 & (g^-)^2 \end{pmatrix}$$

$$+ 4 \begin{pmatrix} (g^+)^2 & g^+(g^+ - g^-)F_d + (g^+ - g^-)F_d g^- \\ 0 & (g^-)^2 \end{pmatrix}$$

With (12.95) being zero, $[\Lambda_d, \Lambda_a] = 0$

Note, a stationary condition at zero evaluates to no current.

Exercise #13: (from exercise) "Stationary relations"
(pg 739)

$$\int \frac{d\epsilon}{2\pi} (F(\epsilon + \omega) - F(\omega)) = \frac{\omega}{\pi} \int \frac{d\epsilon}{\pi} (1 - F(\epsilon - \omega) F(\epsilon))$$

$$= \frac{\omega}{\pi} F_b(\omega)$$

with $F(\epsilon) = \tanh(\epsilon/2T)$

$$F_b = \coth(\omega/2T)$$

$$g \equiv g_L = g_R$$

(12.95) "Fourier Representation"

$$S_{\text{tun}}[\phi] = \int (d\omega) (\bar{c}, -i\bar{s})_{\omega} \begin{pmatrix} 0 & \Sigma^- \\ \Sigma^+ & \Sigma^K \end{pmatrix} \begin{pmatrix} 0 \\ i s \end{pmatrix}_{\omega}$$

$$\Sigma_{\omega}^+ \simeq -\Sigma_{\omega}^- = \frac{i}{2} \sum_{\sigma} g_{\sigma} \int (d\epsilon) (\bar{F}_{d,\sigma} - F_{\sigma,\epsilon-\omega})$$

$$\Sigma_{\omega}^K = i \sum_{\sigma} g_{\sigma} \int (d\epsilon) (1 - F_{d,\epsilon} F_{\sigma,\epsilon-\omega})$$

$$\sum_{\omega}^+ \approx -\sum_{\omega}^- = \frac{i}{2} g \left(\frac{2\omega}{\pi} \right)$$

$$= \frac{ig\omega}{\pi}$$

$$\sum_{\omega}^{IK} = ig \left(\frac{\omega}{\pi} F_b(\omega) \right)$$

$$= \frac{2ig}{\pi} K(\omega)$$

Exercise #14: (12.99) "Second Order Action"

(pg 742)

$$S^{(2)}[\Phi] = \frac{1}{2E_c} \int dt \partial_t \Phi_c \partial_t \Phi_q + \frac{g}{2\pi} \int \frac{d\epsilon}{2\pi} (\Phi_c, \Phi_q)_\omega \begin{pmatrix} 0 & -i\omega \\ i\omega & iK(\omega) \end{pmatrix} \begin{pmatrix} \Phi_c \\ \Phi_q \end{pmatrix}_\omega$$

Green's Function:

$$\begin{pmatrix} 0 & -i\omega \\ i\omega & iK(\omega) \end{pmatrix} = \begin{pmatrix} G^{-1} & T \\ T & G^{-1} \end{pmatrix}$$

where $G^{-1} = iK(\omega)$

$$= i \left(\frac{1}{2} \omega F_b(\omega) + \frac{1}{4} \sum_{S=\pm 1} (\omega + sV) F_b(\omega + sV) \right)$$

IF $V=0$, then $G^{-1} = \frac{1}{2} \omega F_b(\omega)$

Exercise #15: Fourier Transform:

$$n_d(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\epsilon t} d\epsilon$$

$$\approx \frac{1}{2\pi} \int_{-v/2}^{v/2} e^{i\epsilon t} d\epsilon$$

$$\approx \frac{1}{2\pi} \frac{e^{i\epsilon t}}{it} \Big|_{-v/2}^{v/2}$$

$$\approx \frac{e^{ivt/2} - e^{-ivt/2}}{2\pi it}$$

$$\approx \frac{\sin\left(\frac{vt}{2}\right)}{\pi t}$$

$$\approx \frac{i \sin\left(\frac{vt}{2}\right)}{\pi (it)}$$

$$\approx \frac{i \sin\left(\frac{vt}{2}\right)}{\pi t}$$

$$(1-n_d)(t) = 1 - \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\epsilon t} d\epsilon$$

$$\approx 1 - \frac{1}{2\pi} \int_{-vt/2}^{vt/2} e^{i\epsilon t} d\epsilon$$

$$\approx 1 - i \sin\left(\frac{vt}{2}\right) / \pi t$$

Complex Number

Trigonometry :

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Exercise #16: (12.108) "Particle Distribution"

(pg 745)

$$V_E(\epsilon) = V_d \text{Re} \int dt e^{i\epsilon t / \hbar} \left\langle e^{-i \int_{-t/2}^{t/2} dt' U(t')} e^{\frac{i}{2}(\phi_2(-t/2) + \phi_2(t/2))} \right\rangle_{\Phi} (1-n_d)(t)$$

$$= V_d \text{Re} \int dt e^{i\epsilon t / \hbar} \left\langle e^{-i \int_{-t/2}^{t/2} dt' U(t')} \right\rangle_{\Phi} (1-n_d)(t) \quad \text{where } \langle \dots \rangle_{\Phi} = 0$$

Exercise #17: (12.97) "Tunneling Action"

(pg 747)

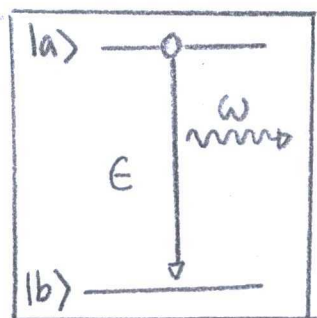
$$S_{\text{tun}}[\Phi] = -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} \left(\Lambda_{\sigma} e^{-i\hat{\Phi}} \Lambda_d e^{i\hat{\Phi}} \right)$$

$$= -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} \left(\Lambda_{\sigma} (1 - i\hat{\Phi} - \frac{\hat{\Phi}^2}{2}) \Lambda_d (1 + i\hat{\Phi} - \frac{\hat{\Phi}^2}{2}) \right)$$

$$= -\frac{i}{4} \sum_{\sigma} g_{\sigma} \text{tr} \left(\Lambda_{\sigma} (1 + iV - \frac{V^2}{2}) \Lambda_d (1 + iV - \frac{V^2}{2}) \right)$$

$\propto |V|$ when real

12.9.1 - Atom-Field Hamiltonian:



a) (12.120) "Model Hamiltonian"

$$H = \frac{\epsilon}{2} \sigma_3 + \sum_k \omega_k a_k^{\dagger} a_k + \sum_k (g_k \sigma_{+} a_k + \bar{g}_k \sigma_{-} a_k^{\dagger})$$

when ϵ = "Energy difference between states"

$|a\rangle = \text{"atomic state"}$

$|b\rangle = \text{"atomic state"}$

$\sigma_i = \text{"Pauli matrices"}$

$\sigma_{\pm} = \text{"Hilbert space"}$

$$= (\sigma_1 \pm i\sigma_2)/2$$

Interaction Energy:

$$\hat{H}_i = \sum_k (g_k \sigma_+ a_k + \bar{g}_k \sigma_- a_k^\dagger)$$

$$= g \sum_k (\sigma_+ a_k + \sigma_- a_k^\dagger)$$

$$= g (\sigma_+ a e^{i\Delta t} + \sigma_- a^\dagger e^{-i\Delta t})$$

where $\Delta = E - \omega$

Master Equation:

(12.4) "Generalized Master Equation"

$$\partial_t \hat{\rho}_s = \hat{L}_s \hat{\rho}_s + \kappa^2 \int_0^t dt' \hat{L}_0 e^{iQLQ\hat{\rho}_s} \hat{L}_i \hat{\rho}_s(t-t')$$

$$= -g^2 \int_0^t dt' e^{i\Delta t} (\langle n+1 \rangle [\sigma_+, \sigma_- \hat{\rho}_s(t-t')])$$

$$- \langle n \rangle [\sigma_+, \hat{\rho}_s(t-t') \sigma_-])$$

$$= g^2 \int_0^t dt' e^{-i\Delta t} (-\langle n+1 \rangle [\sigma_-, \hat{\rho}_s(t-t') \sigma_+])$$

$$+ \langle n \rangle [\sigma_-, \sigma_+ \hat{p}_s(t-t')]]$$

If $\hat{p}_s = p_a P_+ + p_b P_-$ where $P = |a\rangle \text{ or } |b\rangle$
 $1 = p_a + p_b$

then $\partial_t \hat{p}_s = \partial_t p_a + \partial_t p_b$

$$\partial_t p_a = \Gamma (\langle n \rangle p_b - \langle n+1 \rangle p_a)$$

$$\partial_t p_b = \Gamma (-\langle n+1 \rangle p_a + \langle n \rangle p_b)$$

$$\Gamma = 2\pi g^2 \delta(\Delta)$$

(from problem) "Detailed balance Equation"

$$\frac{p_{a,\infty}}{p_{b,\infty}} = \frac{\langle n \rangle}{\langle n+1 \rangle} \approx 1$$

(from problem) "Population imbalance"

$$\Delta p \equiv p_{a,\infty} - p_{b,\infty} = -\frac{1}{2\langle n \rangle + 1}$$

The approximation failed at the master equation by signs, including both positive and negative.

b) Schrödinger Equation:

() "Quantum Mechanical Schrödinger"

$$H\psi = E\psi$$

(part a) "Hamiltonian term for interaction"

$$\hat{H} = g(e^{i\Delta t} \sigma_+ a + e^{-i\Delta t} \sigma_- a^\dagger)$$

(from problem) "Ket"

$$|X, n\rangle \equiv |X\rangle \otimes |n\rangle, \quad X=a, b$$

State #1: $|a, n\rangle$

State #2: $|b, n\rangle$

$$\hat{H}|X, n\rangle = \hat{H}_i |b, n+1\rangle + \hat{H}_i |a, n\rangle$$

$$\hat{H}_i |b, n+1\rangle = g e^{-i\Delta t} (n+1) |a, n\rangle$$

$$\hat{H}_i |a, n\rangle = g e^{i\Delta t} (n+1)^{1/2} |b, n+1\rangle$$

$$|\psi\rangle = \psi_{b, n+1} |b, n+1\rangle + \psi_{a, n} |a, n\rangle$$

() "Time-dependent Schrödinger Equation"

$$i\partial_t \psi = \hat{H}_i(t) \psi$$

$$i\partial_t \psi_{a,n} = g e^{i\Delta t} \sqrt{n+1} \psi_{b,n+1}$$

$$i\partial_t \psi_{b,n+1} = g e^{-i\Delta t} \sqrt{n+1} \psi_{a,n}$$

Method #1: Coupled First-Order Differential

① Matrix forming:

$$i\partial_t \begin{pmatrix} \psi_{a,n} \\ \psi_{b,n+1} \end{pmatrix} = \begin{pmatrix} 0 & g e^{i\Delta t} \sqrt{n+1} \\ g e^{-i\Delta t} \sqrt{n+1} & 0 \end{pmatrix} \begin{pmatrix} \psi_{a,n} \\ \psi_{b,n+1} \end{pmatrix}$$

② Eigenvalues:

$$\begin{pmatrix} -\lambda & g e^{i\Delta t} \sqrt{n+1} \\ g e^{-i\Delta t} \sqrt{n+1} & -\lambda \end{pmatrix} = \lambda^2 - g^2(n+1) = 0$$

$$\lambda_{1,2} = \pm g \sqrt{n+1}$$

③ Eigenvectors:

$$-g \sqrt{n+1} \vec{V}_{11} + g e^{i\Delta t} \sqrt{n+1} \vec{V}_{21} = 0$$

$$g e^{-i\Delta t} \sqrt{n+1} \vec{V}_{11} - g \sqrt{n+1} \vec{V}_{21} = 0$$

$$\begin{aligned} \vec{V}_1 &= \begin{pmatrix} \vec{V}_{11} \\ \vec{V}_{21} \end{pmatrix} \\ &= \begin{pmatrix} e^{+i\Delta t} \\ 1 \end{pmatrix} \end{aligned}$$

$$+g\sqrt{n+1}\vec{V}_{21} + g e^{i\Delta t}\sqrt{n+1}\vec{V}_{22} = 0$$

$$g\sqrt{n+1}\vec{V}_{21} + g e^{i\Delta t}\sqrt{n+1}\vec{V}_{22} = 0$$

$$\begin{aligned}\vec{V}_2 &= \begin{pmatrix} \vec{V}_{21} \\ \vec{V}_{22} \end{pmatrix} \\ &= \begin{pmatrix} -e^{i\Delta t} \\ 1 \end{pmatrix}\end{aligned}$$

④ General Solution:

$$\Psi = C_{1,2} \begin{pmatrix} e^{i\Delta t} \\ 1 \end{pmatrix} e^{ig\sqrt{n+1}t} + C_{3,4} \begin{pmatrix} -e^{i\Delta t} \\ 1 \end{pmatrix} e^{ig\sqrt{n+1}t}$$

$$= \begin{pmatrix} \Psi_{a,n} \\ \Psi_{b,n+1} \end{pmatrix}$$

$$\Psi_{a,n} = C_1 e^{i(\Delta + g\sqrt{n+1})t} - C_2 e^{i(\Delta - g\sqrt{n+1})t}$$

$$\Psi_{b,n+1} = C_3 e^{ig\sqrt{n+1}t} + C_4 e^{-ig\sqrt{n+1}t}$$

④ Exact Solution:

Book had no boundary or initial

conditions. $C_1 = g\sqrt{n+1}$, $C_2 = g\sqrt{n+1}$

$C_3 = i(\Delta + g\sqrt{n+1})$, $C_4 = i(\Delta - g\sqrt{n+1})$ had exact

solutions.

$$\psi_{a,n} = g\sqrt{n+1} \left[e^{i(\Delta+g\sqrt{n+1})t} + e^{-i(\Delta-g\sqrt{n+1})t} \right]$$

$$\psi_{b,n} = i \left[(\Delta+g\sqrt{n+1}) e^{g\sqrt{n+1}t} - (\Delta-g\sqrt{n+1}) e^{-g\sqrt{n+1}t} \right]$$

Method #2: Second-Order Differential:

① Second-Order forming:

$$i \partial_t \psi_{a,n} = g e^{i\Delta t \sqrt{n+1}} \psi_{b,n}$$

$$i \partial_t^2 \psi_{a,n} - g e^{i\Delta t \sqrt{n+1}} \partial_t \psi_{b,n} - i \Delta g e^{i\Delta t \sqrt{n+1}} \psi_{b,n} = 0$$

$$\partial_t^2 \psi_{a,n} + g^2(n+1) \psi_{a,n} - \Delta g e^{i\Delta t \sqrt{n+1}} \psi_{b,n} = 0$$

$$i \partial_t \psi_{b,n} = g e^{-i\Delta t \sqrt{n+1}} \psi_{a,n}$$

⋮

$$\partial_t^2 \psi_{b,n} + g^2(n+1) \psi_{b,n} + \Delta g e^{-i\Delta t \sqrt{n+1}} \psi_{a,n} = 0$$

② Linear Equation of Constant Coefficients:

$$\lambda_a^2 + g^2(n+1) = 0, \lambda_{a1,a2} = \pm i g \sqrt{n+1}$$

$$\lambda_b^2 + g^2(n+1) = 0, \lambda_{b1,b2} = \pm i g \sqrt{n+1}$$

③ Homogeneous Equation:

$$\begin{aligned} \mathcal{U}_{a,n} &= C_1 e^{ig\sqrt{n+1}t} + C_2 e^{-ig\sqrt{n+1}t} \\ \mathcal{U}_{b,n} &= C_3 e^{ig\sqrt{n+1}t} + C_4 e^{-ig\sqrt{n+1}t} \end{aligned}$$

④ Method of Undetermined Coefficients:

$$\mathcal{U}_{a,n} = t^s e^{\alpha t} (R_m(t) \cos \beta t + T_m(t) \sin \beta t)$$

$$s_0, s=0, \alpha=0, \beta=0$$

$$\text{Where } \mathcal{U}_{a,n} = R_m(t)$$

$$\text{Input into } \partial_t^2 \mathcal{U}_{a,n} + g^2(n+1) \mathcal{U}_{a,n} - \Delta g e^{i\Delta t / \sqrt{n+1}} \mathcal{U}_{b,n} = 0$$

$$g^2(n+1) R_m(t) = \Delta g e^{i\Delta t / \sqrt{n+1}} \mathcal{U}_{b,n}$$

$$R_m(t) = \frac{\Delta e^{i\Delta t}}{g\sqrt{n+1}} \mathcal{U}_{b,n}$$

$$\text{Finally } \mathcal{U}_{a,n} = \frac{\Delta e^{i\Delta t}}{g\sqrt{n+1}} \mathcal{U}_{b,n}$$

$$\mathcal{U}_{b,n} = t^s e^{\alpha t} (R_m(t) \cos \beta t + T_m(t) \sin \beta t)$$

$$s_0, s=0, \alpha=0, \beta=0$$

$$\text{Where } \mathcal{U}_{b,n} = R_m(t)$$

$$\text{Input into } \partial_t^2 \mathcal{U}_{b,n} + g^2(n+1) \mathcal{U}_{b,n} - \Delta g e^{i\Delta t / \sqrt{n+1}} \mathcal{U}_{a,n} = 0$$

$$R_m = \frac{-\Delta e^{i\Delta t}}{g\sqrt{n+1}} \mathcal{U}_{a,n}$$

Finally, $\psi_{b,n} = \frac{-\Delta e^{-i\Delta t}}{g\sqrt{(n+1)}} \psi_{a,n}$

⑤ Particular + Equation:

$$\psi_{a,n} = \frac{\Delta e^{i\Delta t}}{g\sqrt{(n+1)}} \psi_{b,n}$$

$$\psi_{b,n} = \frac{-\Delta e^{-i\Delta t}}{g\sqrt{(n+1)}} \psi_{a,n}$$

⑥ Particular + Homogenous:

$$\psi_{a,n} = c_1 e^{+ig\sqrt{(n+1)}t} + c_2 e^{-ig\sqrt{(n+1)}t} + \frac{\Delta e^{i\Delta t}}{g\sqrt{(n+1)}} \psi_{b,n}$$

$$\psi_{b,n} = c_3 e^{ig\sqrt{(n+1)}t} + c_4 e^{-ig\sqrt{(n+1)}t} + \frac{-\Delta e^{-i\Delta t}}{g\sqrt{(n+1)}} \psi_{a,n}$$

⑦ Exact solution:

Impossible! Where are my boundary conditions?

Methods matched, but not to goal without boundaries.

(12.123) "Population Imbalance"

$$\Delta p \equiv \rho_{a, \omega} - \rho_{b, \omega} = \frac{1}{2\langle n \rangle + 1}$$

$$= |\psi_{a,n}|^2 - |\psi_{b,n}|^2$$

$$= \left| e^{i \frac{\Delta}{2} t} \left(\psi_{a,n}(0) \cos(\Omega_n t) - \frac{g_{\text{eff}} \psi_{b,n+1}(0) + \frac{\Delta}{2} \psi_{a,n}(0)}{\Omega_n} \sin(\Omega_n t) \right) \right.$$

$$\left. - e^{-i \frac{\Delta}{2} t} \left(\psi_{b,n+1}(0) \cos(\Omega_n t) - \frac{g_{\text{eff}} \psi_{a,n}(0) - \frac{\Delta}{2} \psi_{b,n+1}(0)}{\Omega_n} \sin(\Omega_n t) \right) \right|$$

$$= \left(1 - \frac{2g^2(n+1)}{\Omega_n^2} \sin^2(\Omega_n t) \right)$$

c) Markovian approximates near zero fluctuations. But far from resonance the system fluctuates with negative population density.

12.9.2 - Weisskopf-Wigner Theory of Spontaneous Emission:

a) (12.120) "Simple model Hamiltonian"

$$\hat{H} = \frac{\epsilon}{2} \sigma_3 + \sum_{\mathbf{k}} \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \sum_{\mathbf{k}} (g_{\mathbf{k}} \sigma_+ a_{\mathbf{k}} + \bar{g}_{\mathbf{k}} \sigma_- a_{\mathbf{k}}^\dagger)$$

(12.126) "Time-dependent Schrödinger Equation"

$$i \partial_t \psi_{a,n} = g e^{i \Delta t} (n+1)^{1/2} \psi_{b,n+1}$$

$$i \partial_t \psi_{b,n+1} = g e^{-i \Delta t} (n+1)^{1/2} \psi_{a,n}$$

Integro-Differential:

$$\partial_t \psi_{a,n} = -i g (n+1)^{1/2} \int_0^t dt' e^{i \Delta t'} \psi_{b,n+1}$$

$$\partial_t \psi_{b,n+1} = -i g (n+1)^{1/2} \int_0^t dt' e^{-i \Delta t'} \psi_{a,n}$$

$$\begin{aligned} b) \partial_t \hat{\rho}_s = & - \int d\omega p(\omega) g^2(\omega) \int_0^t dt' x \left[e^{i(\epsilon-\omega)t'} \langle n(\omega)+1 \rangle \right. \\ & \times [\sigma_+, \sigma_- \rho_s(t-t')] - \langle n(\omega) \rangle [\sigma_+, \hat{\rho}_s(t-t') \sigma_-] \\ & - e^{-i(\epsilon-\omega)t'} (\langle n(\omega)+1 \rangle [\sigma_-, \hat{\rho}_s(t-t') \sigma_+] \\ & \left. + \langle n(\omega) \rangle [\sigma_-, \sigma_+ \rho_s(t-t')]) \right] \end{aligned}$$

as from (12.124)

Problem 12.9.3 - Qubit decoherence from Lindblad
Equation:

12.9.4 - Keldysh theory of single-electron transport:

a) (12.129) "Coulomb blockade"

$$Z = \sum_n \exp\left[-\frac{E_c}{T}(n - N_0)^2\right]$$

(12.93) "Action in the phase-charge representation"

$$\begin{aligned} S[n, \Phi] &\equiv \int dt (n_c \partial_t \Phi_L + n_q \partial_t \Phi_C - 2E_c(n_c + N_0)n_q) + S_{\text{tun}}[\Phi] \\ &= \int dt (n_c \partial_t (\Phi_+ - \Phi_-) + n_q \partial_t (\Phi_+ + \Phi_-) - 2E_c(n_c + N_0)n_q) + S_{\text{tun}}[\Phi] \\ &= \int dt (n_c \partial_t \Phi_+ - n_- \partial_t \Phi_- - E_c(n_+^2 - n_-^2)) + S_{\text{tun}}[\Phi] \end{aligned}$$

(Pg 736) "Tunneling Action"

$$S_{\text{tun}}[\Phi] = 2g \int_{t_1}^{t_2} dt_1 dt_2 (\bar{C} - i\bar{S})_{t_1} \begin{pmatrix} 0 & \Sigma^- \\ \Sigma^+ & \Sigma^k \end{pmatrix}_{t_1-t_2} \begin{pmatrix} C \\ iS \end{pmatrix}$$

$$= 2g \int \frac{d\omega}{2\pi} (\bar{C} - i\bar{S}) \begin{pmatrix} 0 & \Sigma^- \\ \Sigma^+ & \Sigma^k \end{pmatrix} \begin{pmatrix} C \\ iS \end{pmatrix}_{\omega}$$

$$\text{where } C = e^{i\Phi} \cos(\Phi_q/2)$$

$$S = e^{i\Phi} \sin(\Phi_q/2)$$

$$\text{With } L_{\sigma\sigma'} = \frac{1}{4}(\sigma\sigma' \sum^K + \sigma \sum^+ + \sigma' \sum^-)$$

$$\text{and } \sigma\sigma' = \pm 1$$

$$S_{\text{tun}}[\Phi] = \left(\frac{\sqrt{2}}{2}\right) \int dt dt' \begin{pmatrix} e^{-i\Phi_L(t)} & -i\Phi(t) \\ e^{-i\Phi(t)} & \end{pmatrix} L(t-t') \begin{pmatrix} e^{i\Phi_+(t)} \\ e^{i\Phi_-(t)} \end{pmatrix}$$

$$Z = e^{iS_{\text{tun}}[\Phi]}$$

$$= e^{i\left(\frac{\sqrt{2}}{2}\right) \int dt dt' \begin{pmatrix} e^{-i\Phi_L(t)} & -i\Phi(t) \\ e^{-i\Phi(t)} & \end{pmatrix} L(t-t') \begin{pmatrix} e^{i\Phi_+(t)} \\ e^{i\Phi_-(t)} \end{pmatrix}}$$

$$= \sum_{m=-\infty}^{\infty} \frac{\left(\frac{ig}{2}\right)^m}{m!} \cdot \sum_{\{\sigma_k\}} \int D t e^{i \sum_{k=-m}^{2m} \phi_{\sigma_k}(t_k)} \prod_{k=1}^m L_{\sigma_{2k-1}, \sigma_{2k}}(t_{2k-1} - t_{2k})$$

$$= \sum_{m=-\infty}^{\infty} \frac{\left(-\frac{g}{2}\right)^m}{m!} \sum_{\{\sigma_k\}} \int D t e^{-iE_c \int dt (\eta_+^2(t) - \eta_-^2(t))} \prod_{k=1}^m L_{\sigma_{2k-1}, \sigma_{2k}}(t_{2k-1} - t_{2k})$$

b) Temporal range of L :

The temporal range of propagator L scales by t^{-2} from the integral $\int dt e^{iE_c t} \cdot t^{-2}$.

c) (12.130) "Keldysh Integral"

$$Z = \sum_{m=0}^{\infty} \frac{1}{m!} \left(-\frac{g}{2}\right)^m \sum_{\{\sigma_k\}} \int D t e^{-iE_c \int dt (\eta_+^2(t) - \eta_-^2(t))} \prod_{k=1}^m L_{\sigma_{2k-1}, \sigma_{2k}}(t_{2k-1} - t_{2k})$$

Master Equation Derived:

Constraints : $n_0(t) = n$ $n_0(0) = 0$

$$P(n, t_2) = P(n, t_1) + \sum_{n'=n-1}^{n+1} C \cdot P(n', t_1 - \Delta t)$$

Where $C =$ "Transition Probability"

$$= \frac{ig\Delta t}{2} \int dt e^{iE_c t} (n_+^2(t) - n_-^2(t)) \cdot L_+(s)$$

$$= -\frac{g\Delta t}{2E_c} L_+(s)$$

(From problem) "Continuum Approximation"

$$\partial_t P(n, t) = (\Delta t^{-1}) (P(n, t_2) - P(n, t_1 - \Delta t))$$

$$= [(E_+ - 1)W_{n, n+1} - (E_- - 1)W_{n, n-1}] P(n, t)$$

d) Coulomb blockade was from

Chapters 4/5. The Fermi energy

is the 'blockade'. The master

equation; transition probability

changed in time. Specifically,

the description fit rate to a

Circuit. Two rates were measurable,
including $1/LC$ and $1/RC$.

e) Ground state occupancy is in
excess during low voltages.

Too much energy from high
voltage alters rate by $g\nu/4\pi$.

The excited state occurs near
 $g\nu$.